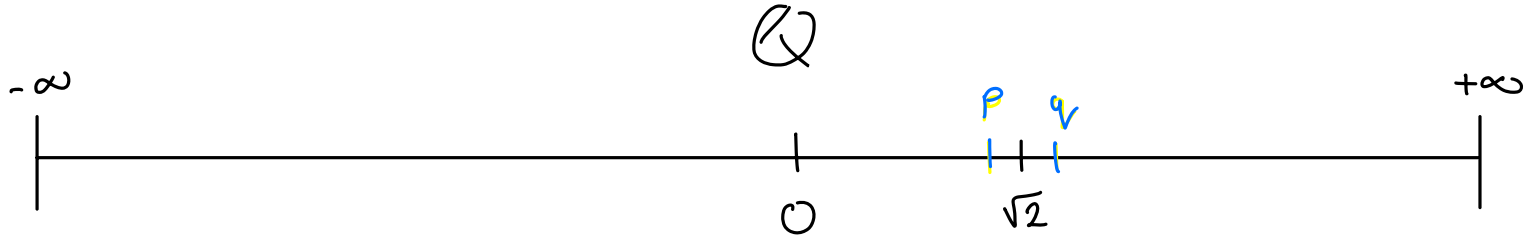
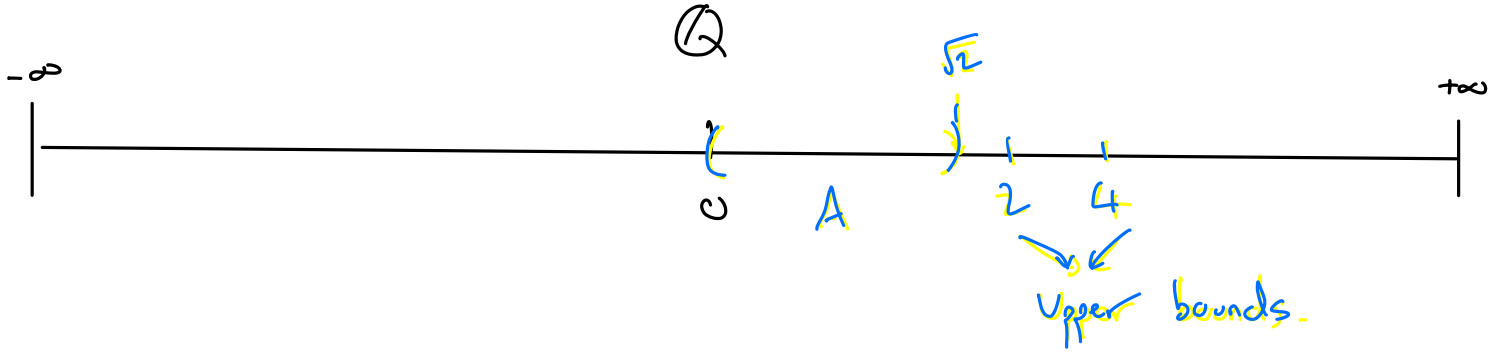
Recap: $\mathbb{Q} \longrightarrow$ Ordered Field $\mathbb{F}$.



$$A = \{ x \in \mathbb{Q} : x > 0,\ x^2 < 2 \} \subset \mathbb{Q}$$



Let A^u be the set of all upper bounds of A. Does A^u have a least upper bound? No.

Outline Proof: Let $p \in A$. Define $q := p - \frac{p^2-2}{p+2}$.

Then $p^2 - 2 < 0$ and $q > p$ as well $q^2 < 2$. That is $q \in A$.

Defn 2.7: Supremum and Infimum

Let A be a non-empty set of an ordered field $\mathbb{F}$ that is bounded above. Then the Supremum of A,

should it exist, is the least upper bound of A, and is denoted by $\sup(A)$. That is, if $a \in \mathbb{F}$ is an upper bound for A, then $\sup(A) \leq a$.

If B is a non-empty subset of an ordered field $\mathbb{F}$, that is bounded below, the infimum of B, should it exist, is the greatest lower bound of B and is denoted by $\inf(B)$. That is, $b \in \mathbb{F}$ is a lower bound for B then $b \leq \inf(B)$.

Defn 2.8: An ordered field $\mathbb{F}$ is said to be complete if for every non-empty subset $A \subset \mathbb{F}$ bounded above, there exists a supremum.

Theorem 2.9: (Real Numbers) There exists a unique complete ordered field $\mathbb{R}$. This field is called the real numbers.

Corollary 2.10: (Real Numbers)

The real numbers are an ordered field with one additional condition/axiom

(i) (Commutative Law)

⋮

(vi) (Order Axioms)

(vii) (Completeness Axiom) Given a non-empty subset $A \subset \mathbb{R}$ that is bounded above, then $\sup(A) \in \mathbb{R}$.

Properties of the Real Numbers

Defn 3.1: (Modulus) The absolute value function $|\cdot| : \mathbb{R} \longrightarrow \mathbb{R}$ is defined as

$$|x| = \begin{cases} x : x \geq 0 \\ -x : x < 0 \end{cases}$$

Ex: $|1| = 1$, $|-2| = 2$, $|0| = 0$.

Theorem 3.3: Properties of Absolute Value Function

(i) $|a| \geq 0$, with equality iff $a = 0$.

(ii) $|a| = |-a|$

(iii) $-|a| \leq a \leq |a|$.

(iv) $|ab| = |a||b|$.

(v) $1/|a| = |1/a|$, if $a \neq 0$.

(vi) $|a|/|b| = |a/b|$, if $b \neq 0$.

(vii) $|a| \leq b$ iff $-b \leq a \leq b$.

(viii) (Triangle Inequality) $|a+b| \leq |a| + |b|$. !!!

(ix) (Reverse Triangle Inequality) $||a|-|b|| \leq |a-b|$.

Proof: (viii) For all $a, b \in \mathbb{R}$ it follows that

$$a + b \leq |a| + |b|.$$

Similarly,

$$-(a+b) = (-a) + (-b) \leq |-a| + |-b| \overset{\text{(ii)}}{=} |a| + |b|$$

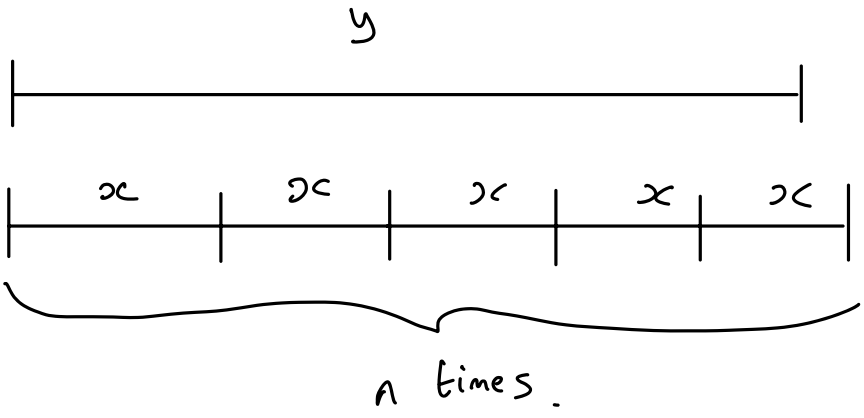
By definition of the absolute value function it follows that

$$|a+b| \leq |a| + |b|. \qquad \square$$

Theorem 3.4 : (Archimedean Property)

For every $x, y \in \mathbb{R}$ with $0 < x < y$, there exists $n \in \mathbb{N}$ such that $y < nx$.

Pic :

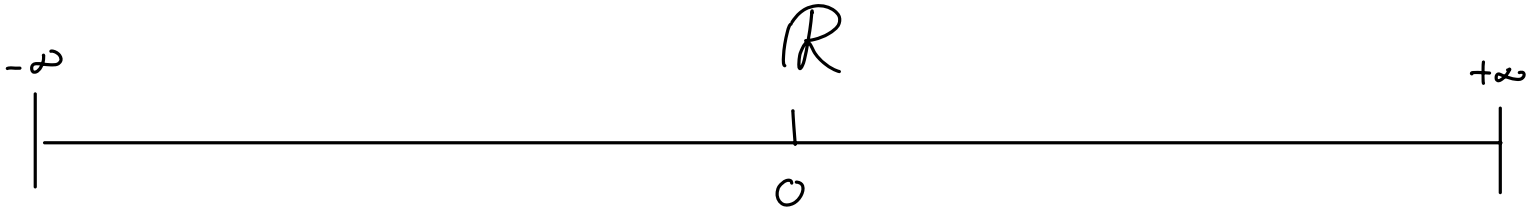


Proof : By setting $z = y/x \in \mathbb{R}$. This means that for every $z \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $z < n$. Assume towards a contradiction that there exists $u \in \mathbb{R}$ such that $n \leq u$ for all $n \in \mathbb{N}$. That is $\mathbb{N}$ is bounded above. Since $\mathbb{N} \subset \mathbb{R}$ by the completeness axiom, $\mathbb{N}$ has a least upper bound, call this $\alpha = \sup(\mathbb{N})$. Then $\alpha - 1 < \alpha$ is not an upper bound for $\mathbb{N}$. Hence, there must exist $m \in \mathbb{N}$ such that

$$\alpha - 1 < m \leq \alpha.$$

$$\Rightarrow \alpha < m + 1 \in \mathbb{N}.$$

This is a contradiction. Hence, $\mathbb{N}$ does not have a least upper bound in $\mathbb{R}$. ☐



Corollary 3.5 : Archimedean Property

For every $z \in \mathbb{R}$ with $z > 0$, there exists $n \in \mathbb{N}$ such that $1/n < z$.

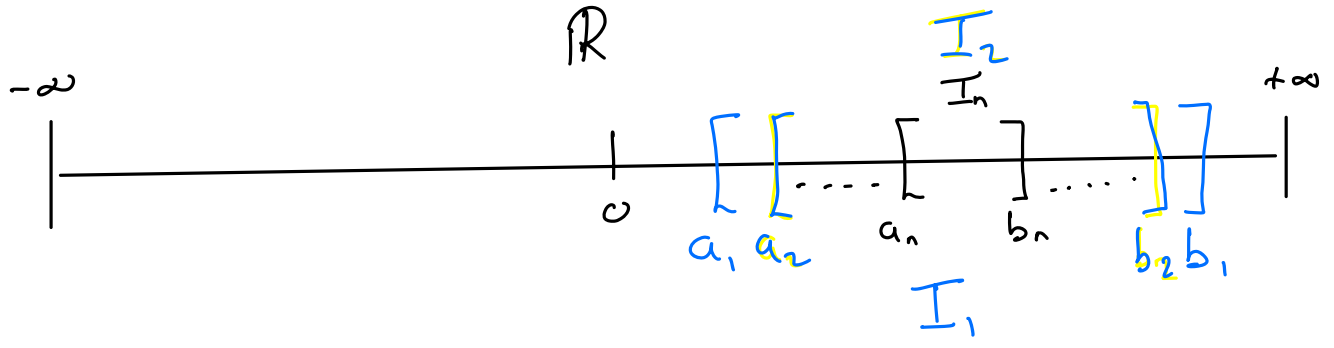
Proof: Follows from Theorem 3.4 by setting $z = x/y$. ☐

Theorem 3.6 : Nested Interval Property

Let $I_n = [a_n, b_n] = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ be a closed interval with $I_{n+1} \subset I_n$ for each $n \in \mathbb{N}$.

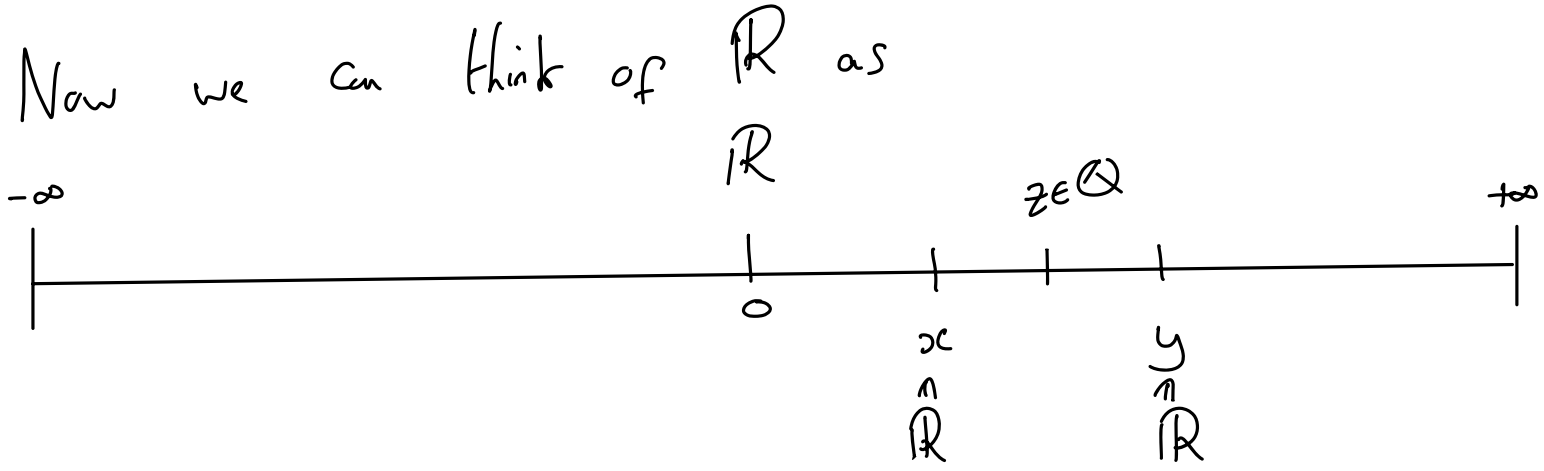
Then there exists an $x \in I_n$ for every $n \in \mathbb{N}$.

Pic:



Proof: Define $A = \{a_n : n \in \mathbb{N}\}$. Since the intervals are nested, that is $I_{n+1} \subset I_n$, it follows that b_n is an upper bound for A. By the completeness axiom, A has a least upper bound, call this α. Hence, $a_n \leq \alpha \leq b_n$ for all $n \in \mathbb{N}$. Then $\alpha \in I_n$ for all $n \in \mathbb{N}$. ☐

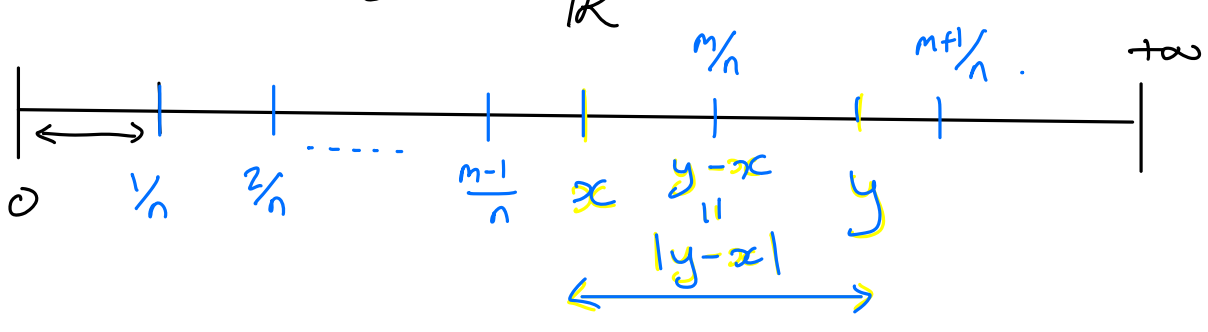
Now we can think of $\mathbb{R}$ as



Theorem 3.7: Density of $\mathbb{Q}$ in $\mathbb{R}$

For every $x, y \in \mathbb{R}$ with $x < y$ there exists $z \in \mathbb{Q}$ such that $x < z < y$.

Pic:



Proof: Suppose $0 < x < y$. Then by Archimedean property there exists an $n \in \mathbb{N}$ such that

$$1/n < y - x.$$

By the definition of $\mathbb{N} \subset \mathbb{R}$, there exists a smallest element $m \in S$ with

$$S = \{ k \in \mathbb{N} : k > nx \}$$

Hence, $nx < m$. Since $m-1 \notin S$ this means $m-1 \leq nx \Rightarrow m \leq nx + 1 < ny$. Putting this together gives

$$x < \frac{m}{n} < y.$$

See notes for other cases. ∎

Corollary 3.8: Density of irrationals in $\mathbb{R}$

For every $x, y \in \mathbb{R}$, with $x < y$ there exists $z \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < z < y$.

Proof: Exercise.

4 - Calculating Suprema and Infima

Theorem 4.1: Uniqueness of Suprema

Let $A \subset \mathbb{R}$ be non-empty and bounded above.

Then $\alpha = \sup(A)$ is unique.

Proof: Assume towards a contradiction that there exists two distinct least upper bounds for A, say $\alpha, \alpha' \in \mathbb{R}$.

Then α is a least upper bound for A so $\alpha \leq \alpha'$.

But α' is also a least upper bound, so $\alpha' \leq \alpha$.

That is $\alpha = \alpha'$. ☐.